INCH-POUND

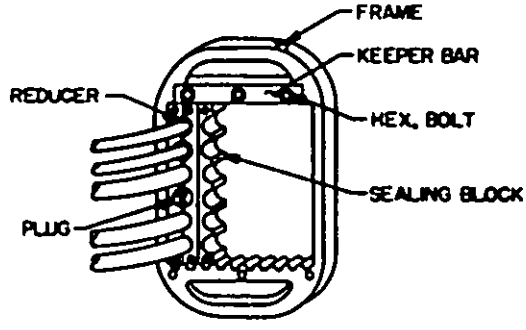
MIL-P-24705/2(SH)
7 June 1990

MILITARY SPECIFICATION SHEET

PENETRATORS, MULTIPLE CABLE, FOR ELECTRIC CABLES, NOTCHED RECTANGULAR METAL FRAME, FOR RECTANGULAR-FACED BLOCKS

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-P-24705.



SH 132316794

FIGURE 1. *Rectangular multiple cable penetrator - assembled view - typical.*

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, decimal tolerance is plus or minus 0.02 (0.5 mm) for dimensions with 2 decimal places or plus or minus 0.016 (0.4 mm) for dimensions with 3 decimal places.
4. Unless otherwise specified, tolerance on steel castings (frames) shall be in accordance with the formula:

   plus or minus toterance = 0.016W1/3 + 0.022L1/3

   where:

   W = Weight of casting (pounds)
   L = Nominal dimension (inch)
5. Unless otherwise specified, tolerances on malleable and ductile iron casting (front and rear compression plates) shall be plus or minus 0.030 inch on the first 3 inches or fraction thereof, plus, plus or minus 0.010 inch for each additional inch.
6. Unless otherwise specified, for molded rubber parts, tolerance shall be in accordance with MIL-HDBK-149.

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Assembly

Components required for each complete notched rectangular metal frame multiple cable penetrator assembly specified by this specification sheet shall be as follows:

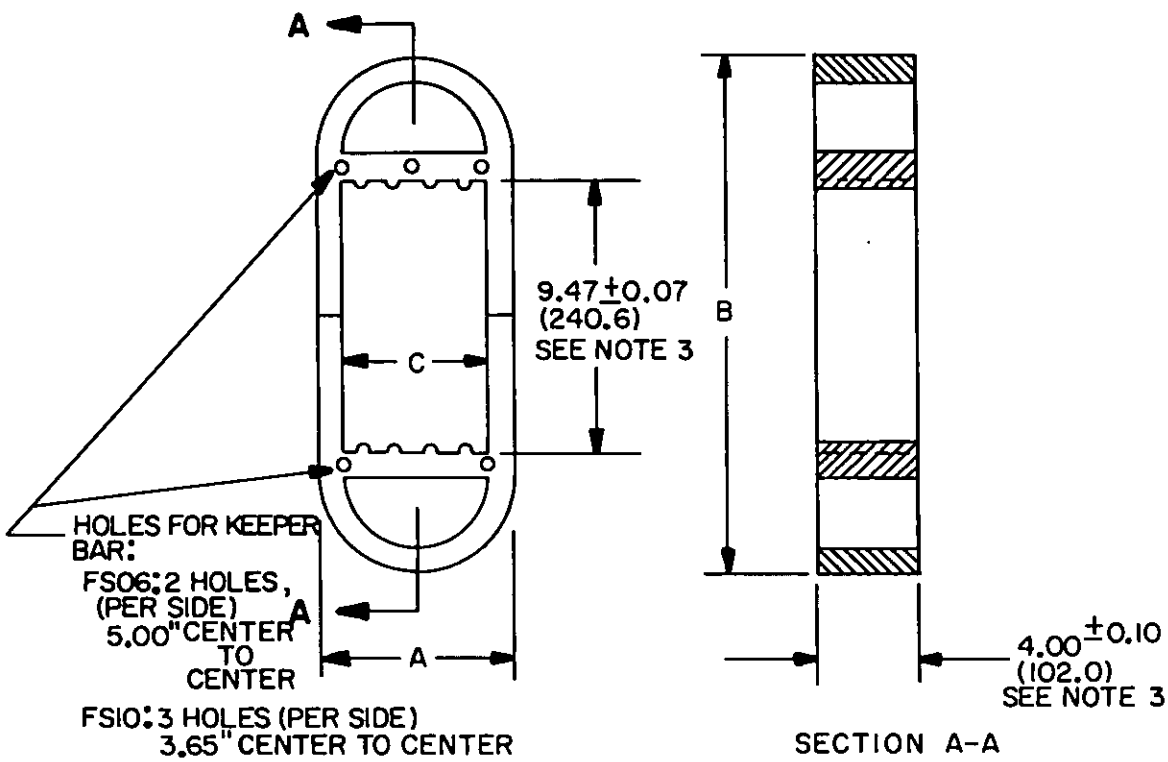
a. One notched rectangular metal frame, including all necessary keeper bars and cap screws (see figures 2, 3, and 4)

b. Rectangular-faced cable insert blocks with associated compression plates and hardware to fill the interior of the rectangular frame (see figures 5, 6, and 7)

c. Blanking insert blocks (see figure 8)

d. Reducers and filler inserts (see figures 9 and 10).

Construction



SH 132316795

FIGURE 2. *Notched rectangular frame - size 06 or 10.*

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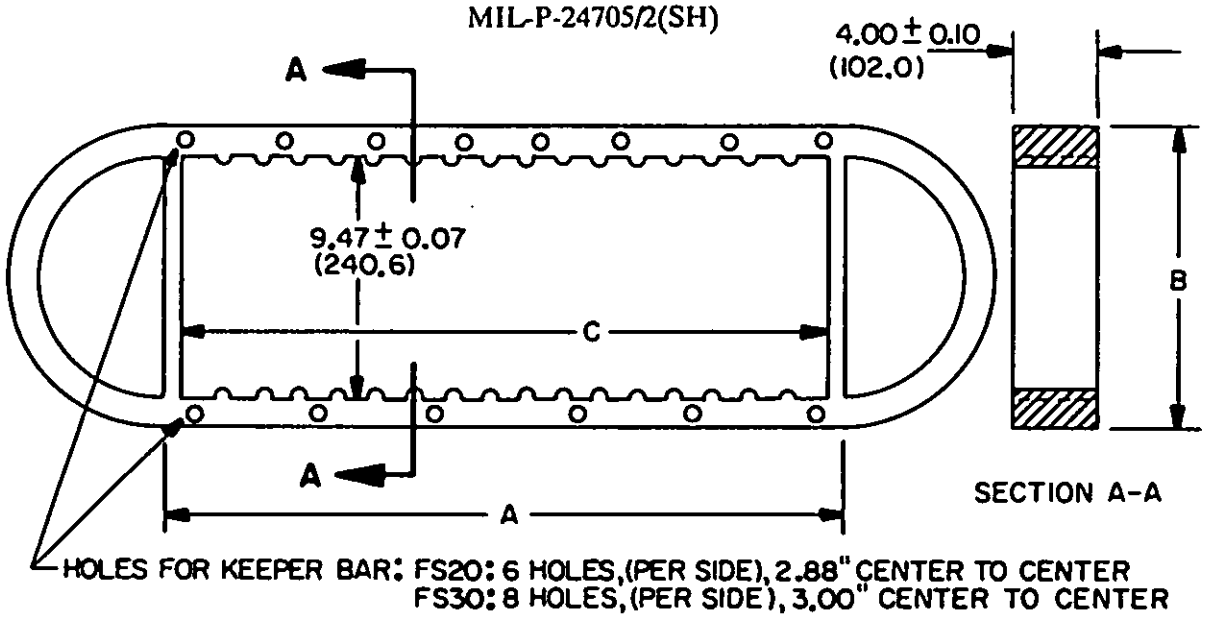


FIGURE 3. *Notched rectangular frame - size 20 or 30.*

TABLE I. *Notched rectangular frame dimensions.*[1]

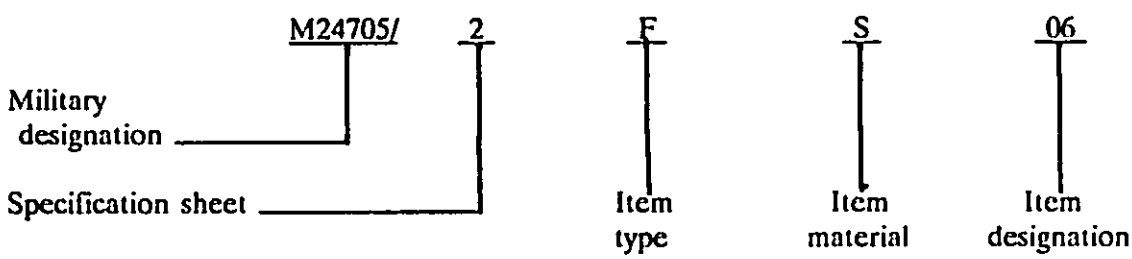
Designator	A	Tolerance (inch)	B	Tolerance (inch)	C	Tolerance (inch)
FS06	5.90 (149.9)	± 0.09	13.00 (330.2)	± 0.11	4.86 (123.4)	± 0.12
FS10	9.11 (231.4)	± 0.10	14.50 (368.2)	± 0.11	8.10 (205.7)	± 0.20
FS20	23.72 (602.5)	± 0.13	11.00 (279.0)	± 0.11	16.20 (411.5)	± 0.40
FS30	31.82 (808.2)	± 0.14	11.50 (292.1)	± 0.12	24.30 (617.2)	± 0.60

[1]Metal for frames shall be in accordance with MIL-P-24705 or ASTM A 27, grade 60 - 30.

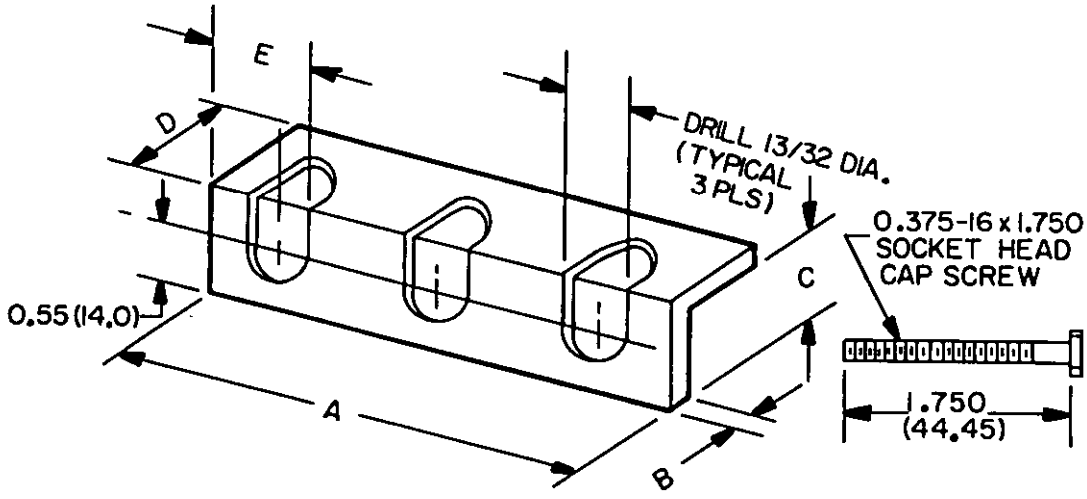
PIN (see MIL-P-24705):

The item designation for notched rectangular frames shall be a two-digit number which corresponds to the number of notches along the A dimension (see figure 2).

EXAMPLE



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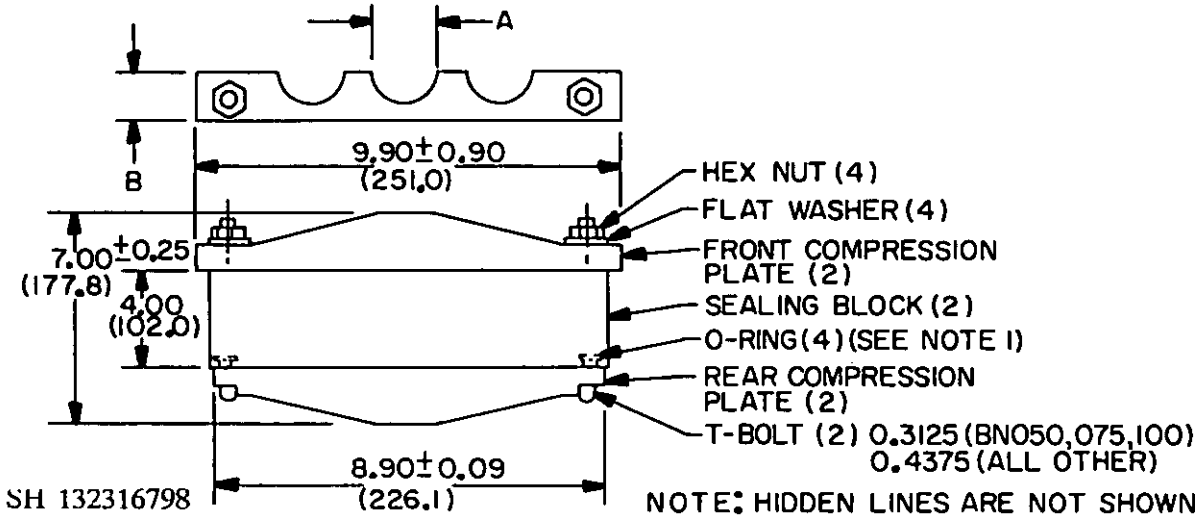
SH 132316797

FIGURE 4. *Keeper bar and cap screw.*

TABLE II. *Keeper bar dimensions.*

Designator[1]	A	B	C	D	E
FS06	5.62 (142.75)	0.12 (3.05)	0.80 (20.32)	1.00 (25.40)	0.31 (7.87)
FS10	7.92 (201.17)	0.12 (3.05)	0.80 (20.32)	1.00 (25.40)	0.31 (7.87)
FS20	15.40 (391.16)	0.19 (4.83)	0.73 (18.54)	1.07 (27.18)	0.50 (12.70)
FS30	23.50 (596.90)	0.25 (6.35)	0.83 (21.08)	1.13 (28.70)	1.25 (31.75)

[1]Metals for keeper bars shall be in accordance with MIL-P-24705, ASTM A 366, or ASTM A 569.
[2]Keeper bars are sized for a specific frame (see table I). Keeper bars shall be supplied with their associated frame.



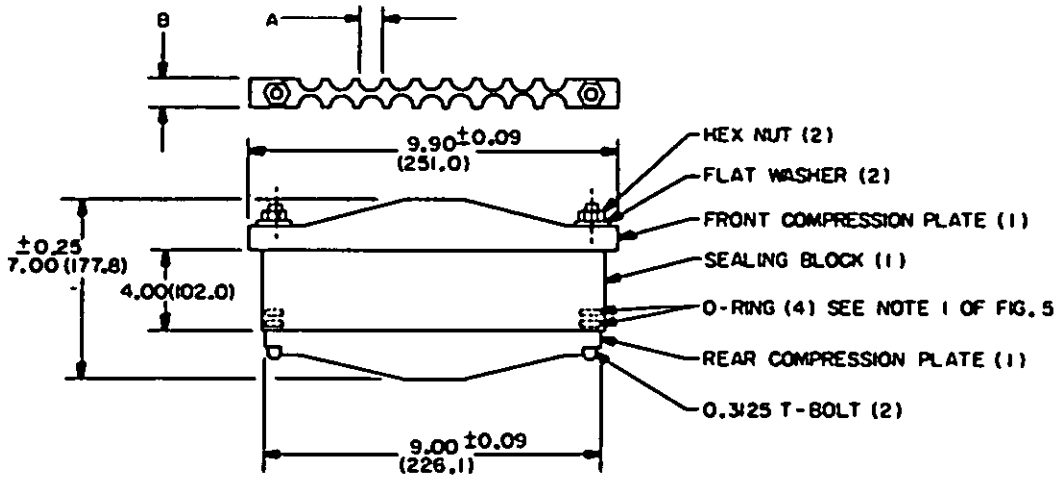
SH 132316798

FIGURE 5. *Standard rectangular-faced cable insert block with associated compression plates and hardware (see requirements).*

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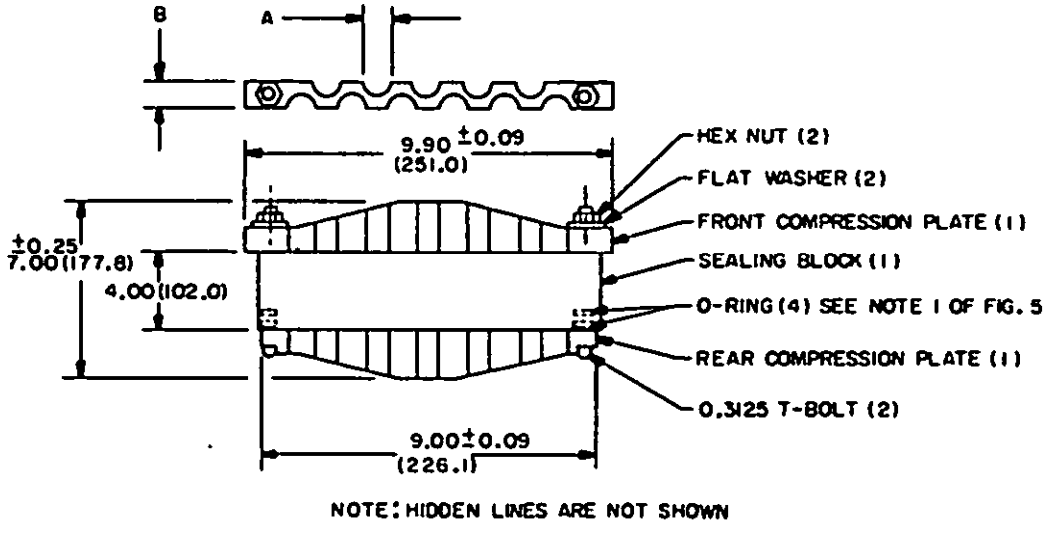
NOTE:

O-rings and lubricant assembled and packaged at factory, shipped wrapped and sealed. DO NOT DISASSEMBLE.



SH 132316799

FIGURE 6. *Special (BN051) rectangular-faced cable insert block with associated compression plates and hardware.*



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FIGURE 7. *Special (BN076) rectangular-faced cable insert block with associated compression plates and hardware.*

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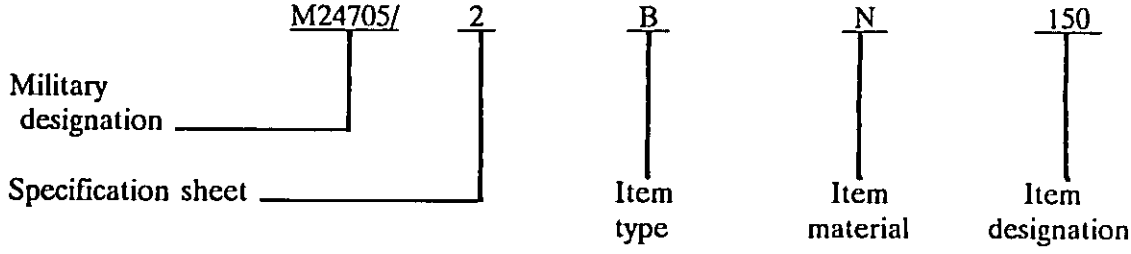
PIN (see MIL-P-24705):

The item designation for rectangular-faced cable insert blocks shall be a three-digit number that represents the passage diameter in inches. The first digit is the whole inch value, and the second and third digits represent the fraction of the inch value in accordance with table III.

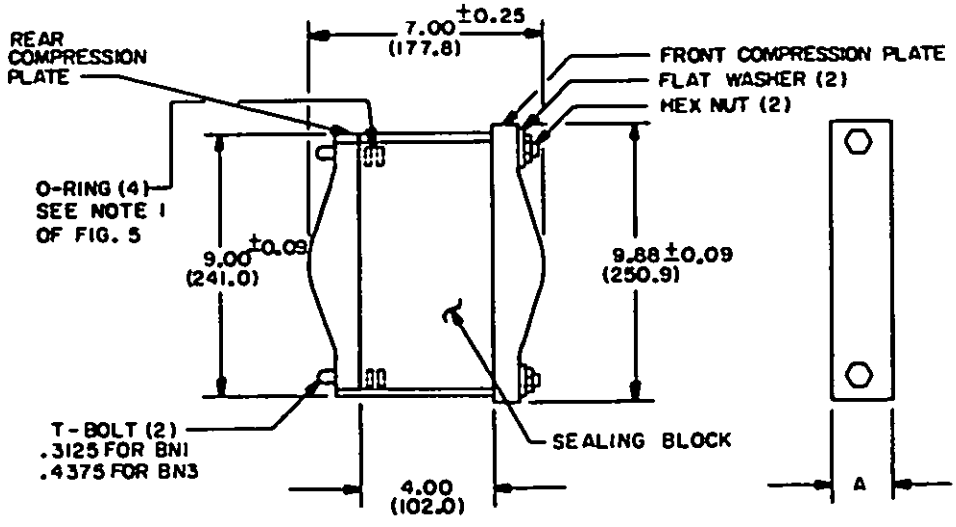
TABLE III. *Rectangular-faced cable insert block dimensions.*

Designator	A	B	Tolerance (inch)	Number of holes
BN050	0.50 (12.7)	0.75 (19.0)	± 0.03	11
BN051	0.50 (12.7)	0.75 (19.0)		11
BN075	0.75 (19.0)	0.75 (19.0)		6
BN076	0.75 (19.0)	0.75 (19.0)		5/6
BN100	1.00 (25.4)	0.75 (19.0)		6
BN125	1.25 (31.8)	1.15 (29.2)		5
BN150	1.50 (38.1)	1.15 (29.2)		4
BN175	1.75 (44.5)	1.15 (29.2)		3
BN200	2.00 (50.8)	1.56 (39.6)		3
BN225	2.25 (57.2)	1.56 (39.6)		3
BN250	2.50 (63.5)	1.97 (50.0)	± 0.035	2
BN275	2.75 (70.0)	1.97 (50.0)		2
BN300	3.00 (76.2)	1.97 (50.0)		2
BN325	3.25 (82.6)	2.38 (60.5)		2
BN350	3.50 (89.0)	2.38 (60.5)		2

EXAMPLE



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SH 132316801

FIGURE 8. *Blanking insert block with associated compression plates and hardware (see requirements).*

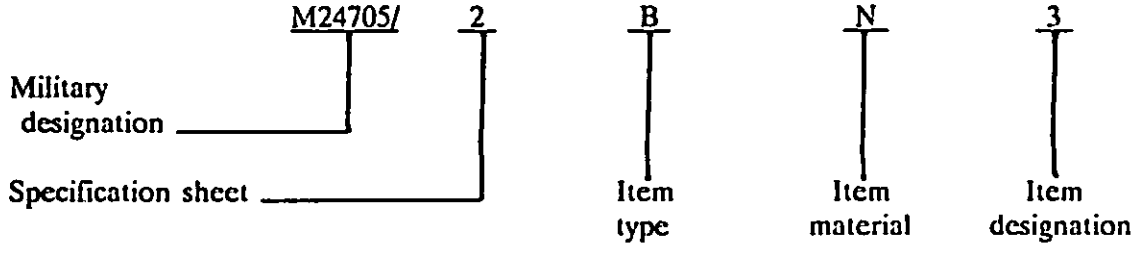
TABLE IV. *Blanking insert block dimensions.*

Designator	A	Tolerance (inch)
BN1	0.75 (19.0)	± 0.03
BN3	2.38 (60.5)	± 0.03

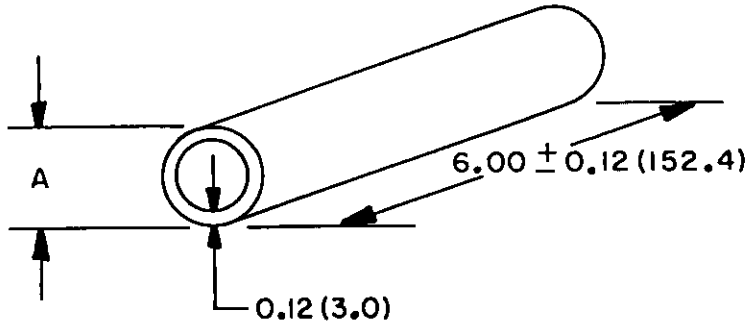
PIN (see MIL-P-24705):

The item designation for blanking insert blocks shall consist of a single-digit number that represents the width of the block by the number of frame notches the block occupies.

EXAMPLE



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SH 132316810

FIGURE 9. *Reducer.*

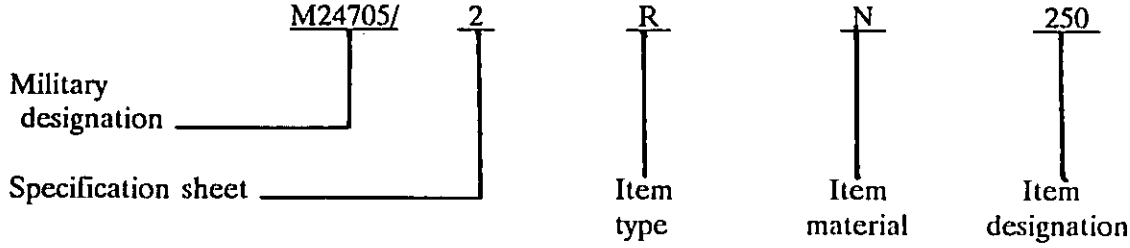
TABLE V. *Reducer dimensions.*

Designator	A	Tolerance (inch)
RN075	0.75 (19.0)	± 0.050
RN100	1.00 (25.4)	± 0.050
RN125	1.25 (31.8)	± 0.063
RN150	1.50 (38.0)	± 0.063
RN175	1.75 (44.5)	± 0.080
RN200	2.00 (50.8)	
RN225	2.25 (57.2)	
RN250	2.50 (63.5)	
RN275	2.75 (70.0)	
RN300	3.00 (76.2)	
RN325	3.25 (82.6)	
RN350	3.50 (89.0)	

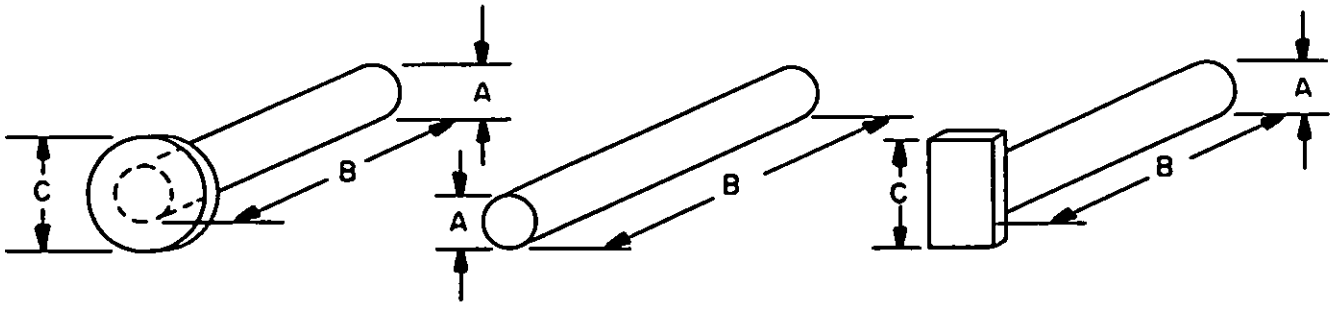
PIN (see MIL-P-24705):

The item designation for reducers shall consist of three-digits that correspond to the rectangular-faced cable insert block with which the reducer is associated.

EXAMPLE



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IN250 AND IN300 ONLY IN150 AND IN200 ONLY IN050, IN100, AND IN350 ONLY

SH 132316811

FIGURE 10. *Filler insert.*

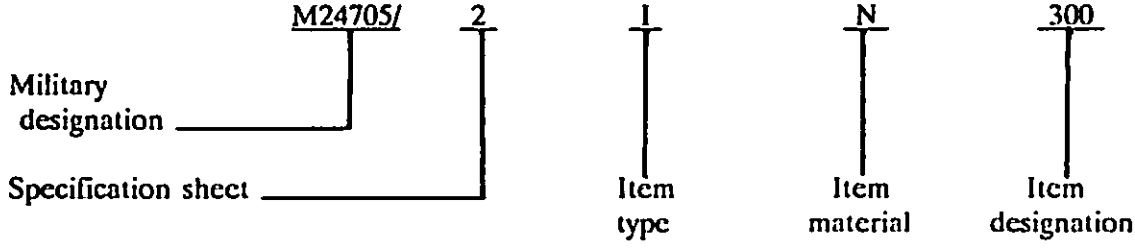
TABLE VI. *Filler insert dimensions.*

Designator	A	Tolerance (inch)	B	Tolerance (inch)	C
IN050	0.50 (12.7)	± 0.005	5.70 (145.0)	± 0.05	0.94 (23.9)
IN100	1.00 (25.4)	± 0.008	5.70 (145.0)		1.38 (35.1)
IN150	1.50 (38.0)	± 0.008	4.00 (102.0)		N/A
IN200	2.00 (50.8)	± 0.010			N/A
IN250	2.50 (63.5)				3.50 (88.9) DIA
IN300	3.00 (76.2)				3.88 (98.6) DIA
IN350	3.50 (89.0)				4.50 (114.3)

PIN (see MIL-P-24705):

The item designation for filler inserts shall consist of three-digits that correspond to the rectangular-faced cable insert block with which the filler insert is associated.

EXAMPLE



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REQUIREMENTS

Nonmagnetic:	When specified (see MIL-P-24705)
Durometer:	62 ± 10
Compression:	12 percent (maximum) method B
Expansion:	1 percent (minimum)
Sample size:	One half of one BN1 (4.00 by 0.75 by 4.50 inch portion)
Fluid immersion:	Plus or minus 5 (maximum)
Sample size:	Three IN150
IR signature:	Informational
Sample size:	One IN150

Keeper bars: Metal shall be in accordance with MIL-P-24705 and ASTM A 131, grade A. The A dimension for each keeper bar (see figure 4) shall be sized to the applicable frame A dimension shown in table I.

Compression plates: Metal shall be in accordance with MIL-P-24705, or grade 32510 in accordance with ASTM A 47, or grade 60-42-10 in accordance with ASTM A 536. The A and B dimensions for each compression plate (see figures 5, 6, and 7) shall be sized to the specific cable insert block shown in table II. The A dimension for each compression plate (see figure 8) shall be sized to the specific blanking insert block shown in table III.

Identification marking: Identification marking shall not be required on the faces of the cable insert blocks or reducers. Passage diameter shall not be required on filler inserts.

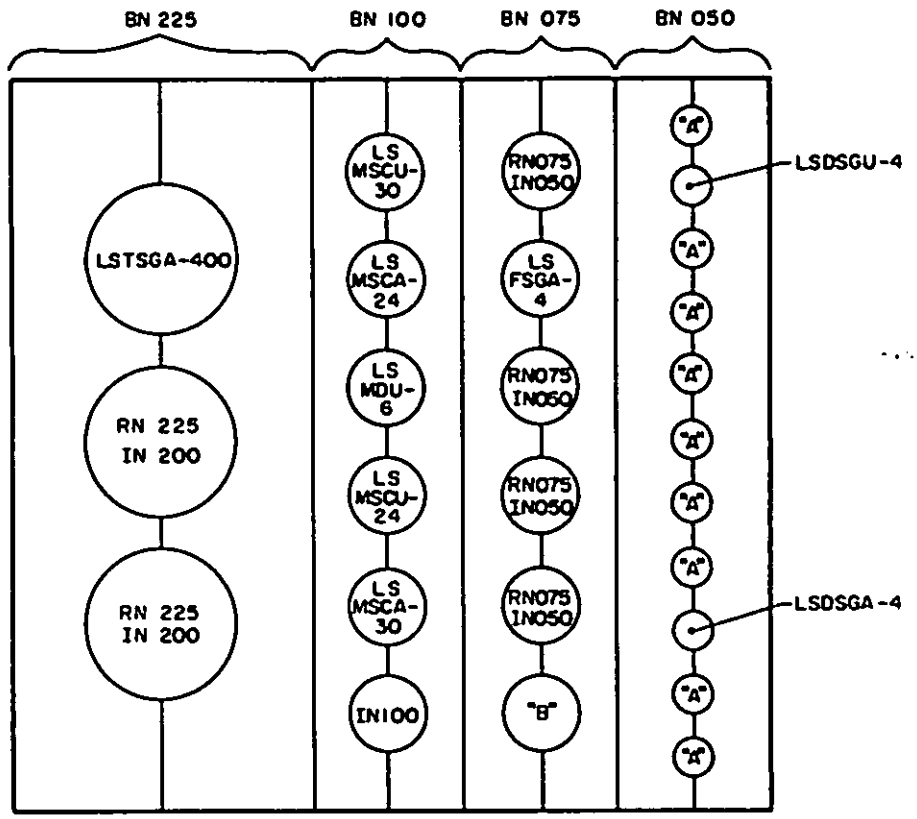
Notched rectangular frame multiple cable penetrator test specimen (see figure 11).

The components shall be as specified. Each component listed shall be required for each test specimen.

Frame:	1 - FS10
Rectangular-faced cable insert blocks with compression plates, bolts, and hex nuts:	2 - BN050 2 - BN075 2 - BN100 2 - BN225
Reducers:	5 - RN075 2 - RN225
Filler inserts:	13 - IN050 1 - IN100 2 - IN200

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Cables:	LSDSGA-4, LSDSGU-4 LSFSGA-4, LSFSGU-4 LSMSCA-24, LSMSCU-24 LSMSCA-30, LSMSCU-30 LSMDU-6, LSTSGA-400
Lubricant:	MIL-S-8660
Compression torque values:	5/16-inch nuts - 16 foot-pounds 7/16-inch nuts - 45 foot-pounds



SH 132316812

"A" = IN050 filler insert plug
"B" = LSFSGU-4 (and RN075 reducer if cable diameter is 0.005 or less)

FIGURE 11. *Notched multiple cable penetrator test specimen - cable configuration.*

Preparing activity:
Navy - SH
(Project 5975-N002-2)

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